

1 Chapter 5

1.1 Discrete Uniform Distribution

$$f(x) = \frac{1}{b - a + 1}, x = a, a + 1, \dots, b$$

1.2 Hypergeometric Distribution

$$f(x) = \frac{\binom{r}{x} \binom{N-r}{n-x}}{\binom{N}{n}}$$

$$\min(r, n) \geq x \geq \max(0, n - (N - r))$$

N objects in total where r are successes, and we randomly select n without replacement

X = number of successes observed among n trials

$$X \sim \text{Hypergeo}(N, r, n)$$

1.3 Binomial Distribution

$$f(x) = \binom{n}{x} p^x (1 - p)^{n-x}, x = 0, 1, 2, \dots, n$$

Requirements:

- Two Trials
- Independent Trials (With replacement)
- Multiple Trials
- Same Probability

X = number of successes observed among n trials

$$X \sim \text{Binom}(n, p)$$

1.4 Negative Binomial Distribution

$$f(x) = \binom{x+k-1}{x} p^k (1-p)^x, x = 0, 1, 2, \dots$$

Two outcomes, independent trials, same probability of success. Do experiments until we obtain k successes. X = the number of **failures** obtained before k^{th} success.

$$X \sim \text{NB}(k, p)$$

1.5 Geometric Distribution

$$f(x) = p(1-p)^x, x = 0, 1, 2, \dots$$

Two outcomes, independent trials, same probability of success. X = the number of **failures** obtained before 1st success.

$$X \sim Geo(p)$$

1.6 Poisson Distribution

$$f(x) = \frac{e^{-\mu} \mu^x}{x!}, x = 0, 1, 2, \dots, \mu > 0$$

X = the number of events of some type. The events occur according to $\mu = np$.

1.7 Poisson Distribution from Poisson Process

Order notation:

$$g(\Delta t) = o(\Delta t), \Delta t \rightarrow 0$$

This means g approaches 0 quicker than Δt as Δt approaches 0.

$$\frac{g(\Delta t)}{\Delta t} \rightarrow 0$$

Setup: assume a certain type of event occurs at random points in time and satisfies the following

- **Independence** The number of occurrences in non-overlapping intervals are independent
- **Individuality** $P(2 \text{ or more events in } (t, t + \Delta t)) = o(\Delta t)$ as $t \rightarrow 0$. (Events do not occur in clusters, but individually)
- **Homogeneity / Uniformity** Occurs at a uniform rate λ over time so

$$P(\text{one event in } (t, t + \Delta t)) = \lambda \Delta t + o(\Delta t)$$

$$f(x) = P(X = x) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}, x = 0, 1, 2, \dots$$

X = number of events in a specified length

$$X \sim Poi(\lambda), \mu = \lambda t$$

Also applies when events occur randomly in space (replace time with volume or area)

2 Chapter 7

2.1 Frequency Distribution

- Mean > Median (right-skewed)
- Mean = Median (Symmetric)
- Mean < Median (Left-Skewed)

2.2 Arithmetic Mean

$$\bar{x} = \sum_{i=1}^n \frac{x_i}{n}$$

2.3 Expected Value

Expected value (mean or expectation) of a **discrete** random variable X with probability function $f(x)$ is:

$$\mu = E(X) = \sum_{all\ x} xf(x)$$

2.4 Properties of Expectation

$$E[ag(X) + b] = aE[g(X)] + b$$

$$E[Ag_1(X) + bg_2(X)] = aE[g_1(X)] + bE[g_2(X)]$$

$$E(a) = a, E[g(X)] = g(E[X]) \text{ only if } g(X) \text{ is a linear function}$$

2.5 Expected Values of Distributions

- Binomial $E(X) = np$
- Poisson $E(X) = \mu = \lambda t$
- Discrete $E(X) = \frac{a+b}{2}$
- Hypergeometric $E(X) = \frac{nr}{N}$
- Negative Binomial $E(X) = \frac{k(1-p)}{p}$
- Geometric $E(X) = \frac{1-p}{p}$

2.6 Interquartile Range (IQR)

$$IQR = Q_3 - Q_1, \tilde{x} = Q_2$$

2.7 Variability

$$E(X - \mu) = E(X) - \mu = 0$$

$$\sigma^2 = \text{Var}(X) = E[(X - \mu)^2]$$

$$\sigma = \sqrt{E[(X - \mu)^2]}$$

$$\text{Var}(X) = E(X^2) - \mu^2 = E[X(X - 1)] + \mu - \mu^2$$

2.8 Variance of Distributions

- Binomial $\text{Var}(X) = np(1 - p)$
- Poisson $\text{Var}(X) = \mu^2$
- Discrete Uniform $\text{Var}(X) = \frac{(b - a + 1)^2 - 1}{12}$
- Hypergeometric $\text{Var}(X) = \frac{nr}{N} \left(1 - \frac{r}{N}\right) \frac{N - n}{N - 1}$
- Negative Binomial $\text{Var}(X) = \frac{k(1 - p)}{p^2}$
- Geometric $\text{Var}(X) = \frac{1 - p}{p^2}$

2.9 Properties of Mean and Variance

Let $Y = aX + b$, $E(a) = a$, $\text{Var}(a) = 0$

1. $\mu_Y = E(Y) = aE(X) + b$
2. $\sigma_Y^2 = \text{Var}(Y) = a^2 \text{Var}(X)$

3 Chapter 8

3.1 Cumulative Distribution Function (c.d.f)

- $F(x) = P(X \leq x)$
- $F(x)$ is defined for all $x \in \mathbb{R}$
- $\lim_{x \rightarrow -\infty} F(x) = 0, \lim_{x \rightarrow \infty} F(x) = 1$
- $F(x)$ is a non decreasing function
- $P(a < X \leq b) = F(b) - F(a) = \int_a^b f(x)dx$
- $P(X = x) = 0, P(a < X < b) = P(a \leq X \leq b) = P(a \leq X < b) = P(a < X \leq b) = F(b) - F(a)$

3.2 Probability Density Function (p.d.f)

Suppose an interval $[x, x + \Delta x]$

$$P(x \leq X \leq x + \Delta x) = F(x + \Delta x) - F(x)$$

$$f(x) = \lim_{\Delta x \rightarrow 0} \frac{P(x \leq X \leq x + \Delta x)}{\Delta x} = \frac{dF(x)}{dx} = \lim_{\Delta x \rightarrow 0} \frac{F(x + \Delta x) - F(x)}{\Delta x}$$

Continuous r.vs.

$$f(x) = \frac{d}{dx} F(x), F(x) = P(X \leq x) = \int_{-\infty}^x f_x(x)dx$$

Discrete r.vs.

$$f(x) = P(X = x) = F(x) - F(x - 1), F(x) = \sum_{w=-\infty}^x f(w)$$

Properties:

- $P(a \leq X \leq b) = F(b) - f(a) = \int_a^b f(x)dx$
- $f(x) \geq 0$
- $\int_{-\infty}^{\infty} f(x)dx = \int_{all x} f(x)dx = 1$
- $F(x) = \int_{-\infty}^x f(u)du$
- $P(x - \frac{\Delta x}{2} \leq X \leq x + \frac{\Delta x}{2}) = F(x + \frac{\Delta x}{2}) - F(x - \frac{\Delta x}{2}) \simeq f(x)\Delta x$

3.3 Expectation, Mean, Variance of Continuous Distribution

$$E(g(X)) = \int_{all\ x} g(x)f(x)dx$$

$$E(x) = \int_{all\ x} xf(x)dx$$

$$\sigma^2 = Var(X) = E(X^2) - E(X)^2$$

3.4 Continuous Uniform Distribution

X is a r.v. taking on values in the interval $[a, b]$ with all subintervals of a fixed length being equally likely

$$X \sim Unif[a, b]$$

The probability density function (p. d. f) must be a constant $f(x) = k$ so

$$f_X(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & otherwise \end{cases}$$

The Cumulative Distribution function

$$F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases}$$

$$E(X) = \frac{b+a}{2}$$

$$Var(X) = \frac{(b-a)^2}{12}$$

3.5 Exponential Distribution

In a Poisson Process for events in time, then the length of time we wait until the first occurrence.

λ = average number of occurrences per unit of time

θ = average waiting time for an occurrence

$$F_x(x) = 1 - e^{-x/\theta}, x > 0$$

$$f_x(x) = \frac{1}{\theta}e^{-x/\theta}, x > 0$$

$$E(x) = \theta = \frac{1}{\lambda}, Var(X) = \theta^2 = \frac{1}{\lambda^2}$$

3.6 Gamma Function

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} = (\alpha - 1)!$$

3.7 Memoryless Property

For exponential Distributions,

$$P(X > c + b | X > b) = P(X > c)$$

3.8 Normal Distribution

A r.v. defined on $(-\infty, \infty)$ has a normal distribution if it has form

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$-\infty < \mu < \infty, \sigma > 0$$

μ shifts distribution along x axis, σ^2 stretches or pulls the distribution

3.9 Finding Probabilities on N(0, 1) tables

Let $X \sim N(\mu, \sigma^2)$

$$Z = \frac{X - \mu}{\sigma}$$

Then $Z \sim (0, 1)$ (use table)

3.10 Gaussian Distribution

$$X \sim G(\mu, \sigma)$$